

Revision History

Version No.	Revision Date	Change References & Summary
1.001	10/08/2023	Initial Draft 1.0
1.1	01/7/2024	

Table of Content

1. INTRODUCTION 3

2. OBJECTIVES 3

3. PROJECT PHASES AND DELIVERABLES..... 5

4. TIMELINE AND MILESTONES: 8

5. EVALUATION CRITERIA/RESPONSE REQUIREMENTS 11

Operational Efficiency: Automated data extraction, transformation, and loading (ETL) processes reduce manual data entry and manipulation efforts, saving time and minimizing errors.

Strategic Planning and Budgeting: Accurate and comprehensive data supports effective strategic planning, budgeting, monitoring and allocation of resources for future growth and initiatives.

Effective Patient Care: Hospital staff can access and utilize a unified dashboard related to patient care, cost of treatment, patient experience and staffing utilization which aiding in care optimization and efficiency.

Unified Reporting and Analysis: Sidra executives, clinical and research staff can access integrated reports and dashboards that provide a holistic view of operations, finances, patient care, and other critical aspects.

Enhanced Performance Measurement: Key performance indicators (KPIs) can be monitored consistently, allowing the hospital to assess its performance and align activities with strategic objectives.

Reduced Data Latency: Real-time or near-real-time data updates allow for faster reporting, making critical information available for time-sensitive decisions.

Self-Service Analytics: Hospital staff can use user-friendly tools to perform ad hoc queries and analyses without needing extensive technical skills, promoting self-service data exploration and analysis.

Data Visualization: Interactive visualizations and dashboards provide intuitive ways to communicate complex data, enabling quicker understanding of trends and insights.

Data Governance and Security: The Data-Lake allows for centralized data governance, ensuring data security, privacy, access control, and compliance with regulatory standards and best practices (i.e. Role Based access control and data encryption).

Data Catalog and Metadata Management: The data-lake can provide a centralized repository to catalog and manage the metadata associated with the stored data.

Long-Term Data Retention: The data-lake provides a secure and accessible repository for historical data, ensuring valuable insights are retained even as ERP systems evolve.

Predictive Analytics, AI, and Machine Learning: The data-lake solution can provide a rich dataset and platforms for training predictive models, AI, and machine learning algorithms, and support data science workloads. It includes tools and frameworks for statistical analysis, machine learning, and AI.

- Azure Data Lake Storage (ADLS): For scalable data storage.
- Azure Synapse Analytics: For data warehousing.
- Azure Data Factory: For data integration and orchestration.
- Azure Purview: for unified data governance
- Deliverable: detailed Data Architecture Design document

3.3 Infrastructure Setup on Azure

▪ Resource Provisioning

Provision and configure the following resources:

- Azure Databricks workspace
- Azure Data Lake Storage
- Azure Synapse Analytics

▪ Network Configuration

Ensure secure and efficient network setup:

- Virtual Network setup
- Network Security Groups
- Private Endpoints for secure data access
- Deliverable: Complete configuration and environment setup sign-off according to architectural design and best practice.

3.4 Data Ingestion and Integration

▪ Data Ingestion

Implement data ingestion pipelines:

- Batch data ingestion using Azure Data Factory
- Real-time data ingestion using Azure Event Hubs and Stream Analytics

▪ Data Integration, Processing and Transformation

- Integrate data through ETL/ELT processes and perform data cleansing and transformation.
- Develop an efficient and optimized ETL/ELT pipeline.

Deliverable: complete data integration and ETL/ELT pipelines.

3.5 Data Quality and Validation

- Implement data quality checks and validation rules to ensure accuracy and consistency.
- Develop processes to identify and resolve data anomalies, duplicates, and discrepancies.
- Establish data profiling and monitoring mechanisms to ensure ongoing data quality.
- Deliverable: Data Quality and Validation Framework Implementation

3.6 Data Storage and Management

- Organize data in Azure Data Lake Storage

- Address any issues or concerns that arise during deployment.
- Deliverable: Production Data Warehouse and Analytics Environment

3.12 Post-Implementation Support

- Provide ongoing technical support to address any system issues, bugs, or enhancements.
- Monitor system performance, data quality, and user interactions.
- Implement updates and modifications based on evolving business requirements.
- Deliverable: Support and Maintenance Documentation

4. Software Licenses

The vendor should be able to develop and implement the solution using a software stack currently available at Sidra Medicine without additional licensing cost. Please refer to the below technology stack available:

- Azure Databricks
- Azure Data Factory
- Azure Synapse Analytics
- Microsoft PowerBI
- Azure Machine Learning or Databricks Mosaic ML
- Other Azure platforms available within Azure Qatar data center.

If additional software licenses are required, it needs to be mentioned clearly with the projected licensing cost.

5. Timeline and Milestones:

The project is estimated to be completed within 10 months following the major activities and deliverables listed in section 3 above.

The vendor should provide a detailed project plan indicating the project management methodology they will follow.

Activity/Deliverable	Delivery Schedule (from start of engagement)
Requirements gathering and analysis	Month 1
Data-Lake / Data-Lakehouse architecture design	Month 1
Infrastructure setup	Month 1 – Month 2
Data ingestion and integration (both batch and real-time).	Month 1 – Month 5
Data processing and transformation	Month 2 – Month 7

7. Sidra Supplied Resources

- Office space complete with furniture (if the vendor's resources are required on-site)
- Access to internet (if the vendor's resources are required on-site)
- Access to existing data servers and dashboards

8. Duration of the Agreement

Vendor shall complete the SERVICES within 10 months from the project kick off date.,

9. Status Meetings and Reports

the Vendor's Agreement Manager, the Sidra Senior or other Program Manager and other appropriate representatives of the Parties shall meet at a mutually agreed site or by teleconference, as requested by Sidra, to discuss the status of the project. During such status meetings, the Parties shall discuss, the status of the project and any difficulties or issues that may exist, and any proposed changes to dates generally and any Due Dates specifically. Vendor shall keep minutes of the meetings and shall issue minutes in draft form to all meeting attendees promptly following each meeting. Following its receipt thereof, the recipient promptly shall review and provide to the drafter any feedback and comments it may have with respect to such minutes, following which the drafter shall incorporate any feedback and comments into the minutes and re-issue the minutes in final form. At least two (2) business days before each such status meeting, Vendor shall present to Sidra a written report of the status of the project. The report shall include a summary in such detail as Sidra shall reasonably request of the accomplishments and difficulties encountered, suggestions and proposed actions for dealing with and resolving any identified difficulties and the anticipated results, and a comprehensive and consolidated log of all outstanding project-related problems identified by Sidra and Vendor that remain to be resolved. Sidra shall have the right to assume that Vendor is not aware of any problems, issues or risks unless Vendor specifically identifies such problems, issues or risks in the project status reports.

10. Pre-Live testing

Vendor shall arrange Pre-live testing ("**Pre-Live Testing**") in accordance with applicable Test Plan to be agreed by the parties. The Test Plan shall specify the relevant Acceptance Criteria including Data Quality and Dashboards performance metrics, and the sequence and the timings to be followed. Sidra shall report to Vendor any defects discovered during Pre-Live Testing, and Vendor promptly shall correct such defects. When Pre-Live Testing has been successfully concluded, Sidra shall so notify Vendor, and the Software thereafter will be ready for Production Use.

11. Production use

Following successful completion of Pre-Live Testing, Sidra will deploy the Software in production. During the Production Use Period, Vendor shall work with Sidra to stabilize, and then optimize, the use of the Software. If any defects are discovered during the

Technical Expertise:

- Deep knowledge of Data-Lake technologies in general and specifically Azure Databricks and Azure Data Factory stack.
 - Proven expertise in developing and deploying large scale Analytics projects.
 - Deep knowledge and proven expertise in Data Architecture and ETL/ELT best practices, implementation, optimization and support.
 - Deep knowledge proven expertise in Data Governance standards and best practices.
 - Experience in visual analytics and dashboards development using PowerBI platform.
 - Vendor should be listed as Microsoft and Databricks official partner (Gold Partner preferred).
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- Proposal should be very clear and self-explanatory with a straightforward, concise description for each component/deliverable as listed in this RFP request.
 - Delivery mode for the project shall be hybrid with on-site technical engineers and resources throughout the project implementation and post go-live support.
 - Proposal should provide all the technical details related to Data Architecture, ETL/ELT technologies and ML platforms.
 - The proposal should clearly define how the platform will be scalable to handle increased data volume and user loads in the future as well as accommodate new data sources and systems.
 - The proposal should define clear metrics that will be used for data processing, storage and retrieval times.
 - The vendor must have done similar implementations in other organizations
 - The vendor must provide proper references and sample reports for such implementations.
 - The vendor must provide all the documentations required as stated in the deliverable section.
 - The vendor must comply with all technical requirements and information security guidelines.

		Design and Prototyping	- Develop system architecture, user interfaces, and data migration plan
			- Create prototypes and conduct proof of concept
		Development and Testing	- Develop core system functionalities on sandbox environment
			- Conduct unit testing and integration testing
		User Acceptance Testing (UAT)	- Prepare test scenarios and conduct UAT sessions
		Deployment and Training	- Prepare deployment plan and deploy system to production
			- Conduct user training sessions
		Documentation and Knowledge Transfer	- Update documentation and knowledge transfer materials
			- Conduct knowledge transfer sessions with stakeholders and support teams
		OTHERS • List of resources (and their cvs) that vendor will utilize/mobilize for the implementation of the project • CV of proposed project manager • Project organizational chart	